Experiment 09- Collaborative Software Development Using Git and GitHub Through the Fork-and-Pull Request Workflow.

1. Aim

To demonstrate collaborative software development using Git and GitHub by forking a repository, creating a feature branch, modifying files, pushing changes, and creating a pull request.

2. Tools Used

- Git
- GitHub
- Visual Studio Code (VS Code)
- Command Prompt / VS Code Terminal
- Repository: Spoon-Knife

3. Procedure

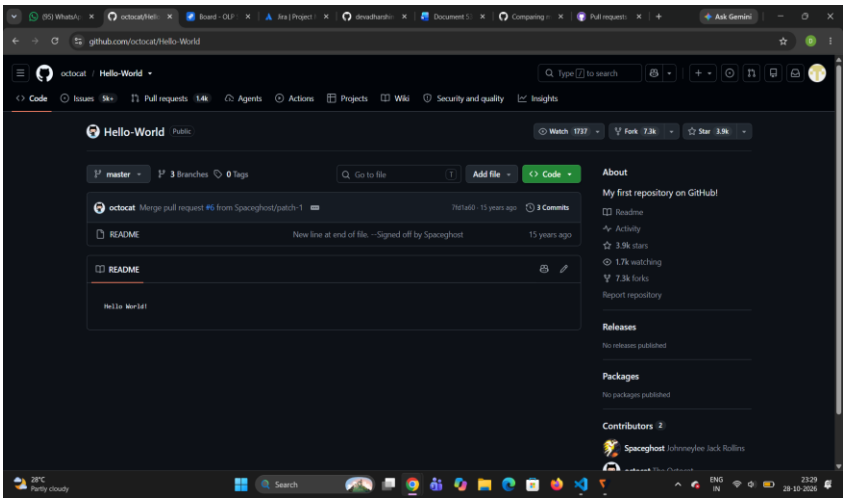
1. Open the public GitHub repository octocat/Spoon-Knife.
2. Clone the repository to the local machine using the git clone command.
3. Create a new feature branch named update-readme.
4. Open the project in VS Code and modify the README.md file.
5. Stage the changes using git add README.md.
6. Commit the changes using git commit -m "Update README documentation".
7. Connect the local project to the GitHub repository using git remote add or git remote set-url.
8. Push the feature branch to GitHub using git push -u myfork update-readme.
9. Open GitHub and create a pull request from update-readme to main.
10. Review the changes, provide feedback, and merge the pull request after approval.

4. Algorithm

1. Start.
2. Open the GitHub repository.
3. Clone the repository to the local machine.
4. Create a feature branch.
5. Modify the README file.
6. Stage and commit the changes.

7. Push the feature branch to GitHub.
8. Create a pull request.
9. Review the changes and provide feedback.
10. Merge the pull request after approval.
11. Stop.

5. Output



6. Result

The Git/GitHub fork-and-pull request workflow was demonstrated successfully. Repository cloning, feature branching, file modification, committing, pushing, and preparing a pull request were performed to understand collaborative software development.